



CENTRAL UNIVERSITY OF SOUTH BIHAR

Department of Computer Science

Class Test - 1: Feb 2026

Programme: M.Sc (CS)

Course Title: Wireless Sensor Networks

Maximum Marks: 10

Course code: CSC82DE01804

Duration: 50:00 Minutes

Semester - 2nd

Note: Questions 1 & 2 are compulsory; answer any one question between 3 and 4.

1. Give a typical block diagram of a sensor node. Why is a sensor node referred to as a transducer? [2+1 Points]
2. List the characteristics of a typical WSN. [3 Points]
3. Discuss any two applications of WSN in wildlife conservation. [4 Points]
4. List any four differences between active and passive sensors. [4 Points]



CENTRAL UNIVERSITY OF SOUTH BIHAR

Department of Computer Science

Class Test - 2: April 2026

Programme: M.Sc (CS)

Course Title: Wireless Sensor Networks

Maximum Marks: 10

Course code: CSC82DE01804

Duration: 50:00 Minutes

Semester - 2nd

Note: Questions 1 & 2 are compulsory; answer any one question between 3 and 4.

1. Define coverage. Discuss its types briefly. [1+2 Points]
2. Explain the neighbor discovery problem and its challenges. [3 Points]
3. What is Clustering? Discuss its advantages. [1+3=4 Points]
4. Explain the flooding algorithm with its advantages and disadvantages. [4 Points]

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CENTRAL UNIVERSITY OF SOUTH BIHAR

Department of Computer Science

Class Test - 3: April 2026

Programme: M.Sc (CS/AI)

Course code: CSC82DE01804/CAI82DE01304

Course Title: Wireless Sensor Networks

Duration: 50:00 Minutes

Maximum Marks: 10

Semester - 2nd

Note: Questions 1 & 2 are compulsory; answer any one question between 3 and 4.

1. Define connectivity. Discuss why good connectivity is important in WSNs. [1+2 Points]
2. Explain localization and its associated challenges. [3 Points]
3. Discuss the cluster formation mechanism used by the LEACH algorithm. [4 Points]
4. Explain the SPIN-BC Algorithm with its advantages and disadvantages. [4 Points]

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दक्षिण बिहार केंद्रीय विश्वविद्यालय, गया

CENTRAL UNIVERSITY OF SOUTH BIHAR

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Question Paper of End –Term Examination May 2026

Name of Department: Deptt. of Computer Science

Programme: M.Sc. (CS)

Session: 2025-2027

Semester: 2nd

Course Title: Wireless Sensor Networks

Course Code: CSC82DE01804

Course Credits: 4

Max. Marks: 70

Duration: 3 Hrs

Enrolment No. (to be filled by Student):.....

Instruction:

1. Attempt questions from all the Sections as directed.

Section - A

Answer all questions below, each carrying 1 point.

(10 X 1 = 10)

1. Define a transducer.
2. What is WSN?
3. List the names of two Optical Sensors.
4. List the types of coverage you can have in a WSN.
5. What is a Flat Wireless Sensor Network?
6. What is a routing hole?
7. What if the probability of transmission p is chosen near 1 in Gossip Routing?
8. Can a neighboring relation be asymmetric? Explain briefly.
9. What do you mean by overhearing?
10. What is eavesdropping?

Section - B

Answer any 5 questions below, each carrying 6 points.

(5 X 6 = 30)

1. What are the main differences between MANET and WSN?
2. What are the active sensors and passive sensors? Discuss with an example.
3. Discuss the virtual force-driven deployment scheme of WSN.
4. What is localization? Why is it needed in WSNs?
5. What are the key challenges of neighbor discovery problem?
6. What are the major sources of energy wasted in Sensor nodes in the context of MAC protocols?

[2+ 4=6 Points]

Section - C

Answer any 2 questions below, each carrying 15 Points.

(2 X 15 = 30)

1. What do you understand by negotiation-based routing protocols? Discuss any two protocols from the SPIN protocol family. [2+ 13=15 Points]
2. Write a short note on the following: [7.5+ 7.5=15 Points]
 - a. Active Sensor Model for Dynamic Sensor Network Programming
 - b. S-MAC protocol
3. Answer the following:
 - a. Discuss the CIA model of Security in the context of WSN. [7.5 Points]
 - b. Discuss engineering and agricultural applications of WSNs in detail. [7.5 Points]

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